PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Quality evaluation of clinical practice guidelines for thromboprophylaxis in orthopedic trauma based on AGREE II and AGREE-REX: a systematic review protocol
AUTHORS	He, Lingxiao; Xie, Jing-ying; Lv, Juan; Liu, Huan; Liao, Deng-bin; Wang, Guang-lin; Ning, Ning; Zhou, Zong-Ke

VERSION 1 – REVIEW

REVIEWER	Prism Schneider University of Calgary, Foothills Medical Centre
REVIEW RETURNED	28-Dec-2021

GENERAL COMMENTS	Overall, this is an important research question and critical appraisal of the current literature is a worthwhile initial step. There are numerous grammatical errors and lay language used throughout that should be corrected. The search terms and timeframe should be carefully considered. Specific recommendations by section are below: Abstract There are a number of grammatical errors in the abstract that should be corrected. For example: thromboembolism(VTE) and clinical practice guidelines(CPGs) and databases(PubMed,…) please add appropriate spacing. Consider changing ethical consent to research ethics board approval, as many studies do not require informed consent from patients, but still require research ethics board approval. - A discussion on the limitations of thromboprophylaxis and risk for bleeding and wound complicaitons, especially in the immediate post-operative period, should be included. Strengths and Limitation “underwent systematic training” should be changed to standardized training for systematic review or systematic review training. Can this be elaborated on? Introduction - Current literature and thrombosis literature tends to use the terminology thromboprophylaxis instead of VTE prophylaxis; therefore, the authors may wish to revise this. - Venous Thromboembolism should be changed to Venous thromboembolism. - The reviewer’s preference is to state patients with DVT, as opposed to DVT patients - DVT and PE should be defined. - Chemical does not need to be capitalized.

	- The authors should ensure that the introduction is written in the same tense throughout. - This statement needs clarification: “it does not fully include the specific conditions of orthopedic trauma patients,” what do the authors mean by specific conditions? - Spacing prior to a cited reference is inconsistent and should be corrected throughout. - This statement needs clarification: “The judgment is not conducive to the choice in clinical practice.” What is meant by “not conducive?” Do the authors mean that what is used in current practice does not agree with CPGs? Patient and Public Involvement - “did not participate” should not be used to refer to interpretation of results, as this is a protocol paper; therefore, the results should not be available at the time of submission of this paper for review. Table 1 - Spacing should be consistent throughout Data Sources - UpTodate should be Up ToDate - a clear statement that grey literature will not be included should be made Search Strategies - The reviewer believes that the following terms should also be included: deep vein thrombosis, thrombosis, thromboprophylaxis, VTE prophylaxis, chemoprophylaxis, chemical prophylaxis, mechanical prophylaxis Data Extraction - Duration of recommended thromboprophylaxis should specifically be reported as well. - It should be clearly stated if the CPGs refer to only operatively treated trauma patients or not. - The authors may want to limit the search to literature within a certain timeframe, as they highlight that inclusion of older literature may be a limitation. QUALITY ASSESSMENT OF CPGS - Spacing prior to brackets is again inconsistent throughout and should be corrected. - Numbers under 10 should be written out in full. - Will there be overlap between the five reviewers conducting quality appraisals and the two reviewers who extract CPG recommendations? - When recommendations conflict with each other, these recommendations will be highlighted. Change to: the discrepancies will be highlighted. Data Management - The tense in this section suggests that the analysis has already been completed – please keep tense consistent throughout the manuscript. - For each of AGREE II and AGREE-REX In the field… - what is meant by “In the field?” - Other than descriptive statistics and ICC, there is limited statistical analysis proposed Discussion

	- Grammatical errors and lay language throughout, this section may benefit from further review - Identifying high-quality recommendations will help us choose an appropriate way to intervene target population – should this be to treat the target population? Or to prescribe appropriate thromboprophylaxis in the trauma population? - The authors could expand on what is meant by the discrepancy between CPG recommendations and the "actual development process," to clarify what is meant and why this may be important.

REVIEWER	Nathan O'Hara University of Maryland
REVIEW RETURNED	21-Feb-2022

GENERAL COMMENTS	The authors plan to perform a systematic review of clinical practice guidelines for venous thromboembolism prophylaxis in orthopaedic trauma patients. The quality of the identified clinical practices guidelines will be assessed using AGREE-II and AGREE-REX. I agree that this information will be relevant to health providers caring for orthopaedic trauma patients and fills a knowledge gap in the field. However, I have a number of concerns with the manuscript in its current form. 1. There are a few inconsistencies with the PROSPERO registration and the manuscript, including: • Start date (June 2006) in PROSPERO vs. inception in manuscript • Three defined research questions in PROSPERO vs. none in the manuscript 2. The authors describe how they will evaluate agreement between reviewers. However, there is limited information on what results will be obtained from the systematic. For example, do the authors plan to identify variation in guidelines, investigate conflicting results, and inform areas for future research? Do they plan to produce statements to guide practice and future guidelines? 3. Many of the current guidelines are written for trauma patients or orthopaedic patients, with orthopaedic trauma patients being a subset of either category. Will these guidelines be included? I also have a few minor comments. 1. The authors state they will review guidelines from inception to Dec 31, 2021. Will guidelines that are published many decades ago still be relevant? 2. Please define each acronym in its first use—for example, ACCP in line 72.

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer 1

Reviewer's Comment	Author Response to Comment	Changes made to the article	Line Number
Abstract			

There are a number of grammatical errors in the abstract that should be corrected. For example: thromboembolism(VTE) and clinical practice guidelines(CPGs) and databases(PubMed,…) please add appropriate spacing.	Thank you for the comment. We have revised as you suggested.	The corresponding modifications are highlighted in gray in the manuscript.	/
Consider changing ethical consent to research ethics board approval, as many studies do not require informed consent from patients, but still require research ethics board approval.	Thanks for your comment. We have added the ethics approval status and approval number.	**Ethics and dissemination** This study will follow the Declaration of Helsinki and has received approval from the Ethics Committee on Biomedical Research, West China Hospital, Sichuan University (Ethics Approval No. 2021-989).	36-38, 269-274
A discussion on the limitations of thromboprophylaxis and risk for bleeding and wound complications, especially in the immediate post-operative period, should be included.	Thanks for your suggestion. We have supplemented the discussion with the possible bleeding complications of chemoprophylaxis. We did not add it in the abstract because the protocol abstract in BMJ OPEN does not contain a discussion.	The use of anticoagulants may increase the risk of bleeding and wound complications in trauma patients, especially in the early postoperative period.	224-226
Strengths and Limitation			
"underwent systematic training" should be changed to standardized training for systematic review or systematic	Thank you for your suggestion. We have modified it. The training received by the researchers is further described in the manuscript. See the section on quality	All the members will not participate in the quality appraisal until received systematic review training. Five reviewers (LH, JX, JL, HL and NN) conducted quality appraisals after systematically learning how to use	52-53, 187-188

review training. Can this be elaborated on?	assessment.	AGREE-II and AGREE-REX.	
Introduction			
Current literature and thrombosis literature tends to use the terminology thromboprophylaxis instead of VTE prophylaxis; therefore, the authors may wish to revise this.	Thanks for your suggestion. We have made changes to the title, keywords, and manuscript.	/	1-3, 42-43, 73
Venous Thromboembolism should be changed to Venous thromboembolism.	Thanks for your comment. We have rechecked this article for grammatical errors and made corrections.	The corresponding modifications are highlighted in gray in the manuscript.	/
The reviewer's preference is to state patients with DVT, as opposed to DVT patients.			
DVT and PE should be defined.			
Chemical does not need to be capitalized.			
The authors should ensure that the introduction is written in the same tense throughout.			
This statement needs clarification: "it does not fully include the specific conditions of orthopedic trauma patients," what do the authors mean by specific conditions?	Thank you for your suggestion. We have added a discussion of the content scope of the ACCP Guideline.	At present, the most widely used VTE prevention clinical practice guideline (CPG) in orthopedic surgery is the 9th edition of the VTE prevention guideline presented by the American College of Chest Physicians (ACCP) in 2012. However, its recommendations for patients with orthopedic trauma are insufficient, covering only patients with hip fractures, isolated lower leg injuries, and knee arthroscopy. And no	78-81

		specific recommendations for the dose and frequency of chemoprophylaxis are mentioned in the above ACCP guideline.	
Spacing prior to a cited reference is inconsistent and should be corrected throughout.	Thank you for raising this point. We have revised as you suggested.	/	/
This statement needs clarification: "The judgment is not conducive to the choice in clinical practice." What is meant by "not conducive?" Do the authors mean that what is used in current practice does not agree with CPGs?	Thanks for your comment. We want to express that the quality of the guideline is not the same as the applicability of the recommendations, so the selection of high-quality guidelines cannot guarantee that the recommendations can be directly applied to clinical practice. We have revised this sentence.	Currently, the Appraisal of Guidelines for Research & Evaluation II instrument (AGREE-II) is widely accepted to evaluate the methodological quality of the guidelines. However, this tool does not assess the specific recommendations provided in the guidelines and cannot guarantee their applicability in clinical practice.	90-91
Patient and Public Involvement			
"did not participate" should not be used to refer to interpretation of results, as this is a protocol paper; therefore, the results should not be available at the time of submission of this paper for review.	Thank you for your comment. We have revised as you suggested.	**Patient and public involvement** Patients and/or the public will not be involved in the design, conduct, reporting, or dissemination plans of this research.	117-119
Table 1			
Spacing should be consistent throughout.	Thank you for your comment. We have revised as you suggested.	/	/

Data Sources			
UpTodate should be Up ToDate. a clear statement that grey literature will not be included should be made.	Thank you for your comment. We have revised as you suggested.	**Eligibility criteria** See details in Table 1. We will only include openly published literature in the study and gray literature will be excluded.	122-123, 140
Search Strategies			
The reviewer believes that the following terms should also be included: deep vein thrombosis, thrombosis, thromboprophylaxis, VTE prophylaxis, chemoprophylaxis, chemical prophylaxis, mechanical prophylaxis.	Thanks for your suggestion. We have integrated your suggestions and revised the search terms to improve the accuracy of the search. See Table 1 for sample strategy, and appendix 2 for detailed search strategies.	Terms related to "orthopedic trauma", "VTE", "prevention" and "guideline" will be adopted for the search in each database. Detailed search strategies are listed in appendix 2. An example of a database search strategy is shown in Box 1.	143-146
Data Extraction			
Duration of recommended thromboprophylaxis should specifically be reported as well. It should be clearly stated if the CPGs refer to only operatively treated trauma patients or not.	Thanks for your comment. We have revised as you suggested.	We will extract the following information of CPGs: (1) title; (2) year of publication; (3) authors; (4) country of development; (5) objective of CPG; (6) target users; (7) method used to collect the evidence; (8) grading methods for the evidence; (9) recommendation formulation method; (10) guideline validation methods; (11) intended users; (12) the specific content of the recommendations given, including the method of thromboprophylaxis, the timing of initiation, dose and duration, and target patients. In addition, we will record each score of items in AGREE-II and AGREE-REX with their reasons.	156-162
The authors may want to limit the search to literature within a certain timeframe, as they	Thanks for your suggestion. We intend to include guidelines for all periods and carefully	**STRENGTHS AND LIMITATIONS OF THIS STUDY** • This study might be included in guidelines that were developed earlier or not updated in time.	56-58, 248-250

highlight that inclusion of older literature may be a limitation.	interpret the results. We have revised our discussion of study limitations.	Thus, more cautious interpretations of the study results should be needed. **DISCUSSION** Our research may still have shortcomings. …Third, to obtain as comprehensive information as possible, we will not limit the start time for the publication of the guidelines. Those recommendations from earlier CPGs may not be consistent with current evidence.	
Quality Assessment of CPGs			
Spacing prior to brackets is again inconsistent throughout and should be corrected. Numbers under 10 should be written out in full.	Thank you for your comment. We have revised as you suggested.	/	/
Will there be overlap between the five reviewers conducting quality appraisals and the two reviewers who extract CPG recommendations?	Thanks for your kind reminder. We have added the name of the reviewers. There will be no reviewer overlap in our quality appraisal and recommendation extracting process.	**QUALITY ASSESSMENT OF CPGS** Five reviewers (LH, JX, JL, HL and NN) conducted quality appraisals after systematically learning how to use AGREE-II and AGREE-REX. **DESCRIPTION AND COMPARISON OF THE RECOMMENDATIONS** We plan to compare the recommendations included in the CPG, which will be extracted independently by two reviewers (DL and GL).	187-188, 193-195
When recommendations conflict with each other, these recommendations will be highlighted. Change to: the discrepancies will be highlighted.	Thanks for your comment. We have revised as you suggested.	Discrepancies will be highlighted when there are conflicts between some recommendations.	200-201
Data Management			

The tense in this section suggests that the analysis has already been completed – please keep tense consistent throughout the manuscript.	Thanks for your comment. We have revised as you suggested.	/	203-219
For each of AGREE II and AGREE-REX In the field… - what is meant by "In the field?" Other than descriptive statistics and ICC, there is limited statistical analysis proposed.	Thanks for your comment. It's a grammar error, and we have revised it. We have added the scoring calculation methods for AGREE-II and AGREE-REX.	For each domain of AGREE II and AGREE-REX, We will calculate the score by the following formula: $$\textit{The scaled domain score} = \frac{Obtained\ score - Minimum\ possible\ score}{Maximun\ possible\ score - Minimum\ possible\ score}$$ Scores for guidelines and recommendations will be described using the mean (SD) and median (IQR) (if needed).	206-210
Discussion			
Grammatical errors and lay language throughout, this section may benefit from further review.	Thanks for your reminder. We have carefully revised the language by a native speaker.	/	/
Identifying high-quality recommendations will help us choose an appropriate way to intervene target population – should this be to treat the target population? Or to prescribe appropriate thromboprophylaxis in the trauma population?	Thanks for your comment. We have revised as you suggested.	Identifying high-quality recommendations will help us to prescribe appropriate thromboprophylaxis in the trauma population.	239-240
The authors could expand on what is meant by the discrepancy	Thanks for your suggestion. We deleted this sentence after careful thought.	~~Finally, we are scoring the CPGs based on the content shown in the guideline document, which may not be consistent with the actual development~~	251-252

between CPG recommendations and the "actual development process," to clarify what is meant and why this may be important.	We meant to say that the methodological quality may be higher than that shown in the guideline text due to irregularity in guideline reporting.	~~process.~~	

Table 2. Response to Reviewer 2

Reviewer's Comment	Author Response to Comment	Changes made to the article	Line
There are a few inconsistencies with the PROSPERO registration and the manuscript, including: • Start date (June 2006) in PROSPERO vs. inception in manuscript • Three defined research questions in PROSPERO vs. none in the manuscript	Thank you for raising this point. The research question in our article is in the objectives section (lines 101-110). In addition, we revised the search timeframe after discussion. Considering that it is June 2022, we reset the search time from inception to Oct 2022. We will submit this update to PROSPERO for consistency.	**Table 1 Inclusion and exclusion criteria.** **Inclusion criteria** (4) Guidelines published from inception to Oct 2022;	124
The authors describe how they will evaluate agreement between reviewers. However, there is limited information on what results will be obtained from the systematic. For example, do the authors plan to identify variation in guidelines, investigate conflicting results, and inform areas for future	Thank you for your suggestion. We have added the information list extracted from the guidelines to provide adequate information.	We will extract the following information of CPGs: (1) title; (2) year of publication; (3) authors; (4) country of development; (5) objective of CPG; (6) target users; (7) method used to collect the evidence; (8) grading methods for the evidence; (9) recommendation formulation method; (10) guideline validation methods; (11) intended users; (12) the specific content of the recommendations given, including the method of thromboprophylaxis, the timing of initiation, dose and duration, and target patients. In addition, we will record each score of	157-164

research? Do they plan to produce statements to guide practice and future guidelines?		items in AGREE-II and AGREE-REX with their reasons.	
Many of the current guidelines are written for trauma patients or orthopaedic patients, with orthopaedic trauma patients being a subset of either category. Will these guidelines be included?	Thanks for your comment. According to our inclusion criteria, we will include any guidelines that have recommendations for orthopedic trauma patients. These recommendations may be distributed among different guidelines for orthopedic surgery, or guidelines for trauma patients.	/	/
The authors state they will review guidelines from inception to Dec 31, 2021. Will guidelines that are published many decades ago still be relevant?	Thanks for your reminder. We have revised the search timeframe and requested an update on PROSPERO. We plan not to set too many restrictions on the publication time of the guideline, to collect relevant literature as much as possible, and we will consider the impact of the guideline publication time when we include literature and recommendations later. This is illustrated in our study limitations.	The following databases and websites from inception to 31 October 2022 will be searched: … **STRENGTHS AND LIMITATIONS OF THIS STUDY** • This study might be included in guidelines that were developed earlier or not updated in time. Thus, more cautious interpretations of the study results should be needed.	128-129 56-58
Please define each acronym in its first use—for example, ACCP in line 72.	Thanks for your comment. We have revised as you suggested.	At present, the most widely used VTE prevention clinical practice guideline (CPG) in orthopedic surgery is the 9th edition of the VTE prevention guideline presented by the American College of Chest Physicians (ACCP) in 2012.	77-78